

HW 9 con. $e_2 = \text{check column 2:}$

C16 Answer check con.: $A \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$
 $= \begin{bmatrix} 1 \cdot 1 + 0 \cdot 1 + 1 \cdot (-1) \\ 1 \cdot 1 + 1 \cdot 1 + 1 \cdot (-1) \\ 2 \cdot 1 + (-1) \cdot 1 + 1 \cdot (-1) \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \checkmark \text{ correct}$

$e_3 = \text{check column 3: } A \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$
 $= \begin{bmatrix} 1 \cdot 1 + 0 \cdot 0 + 1 \cdot (-1) \\ 1 \cdot 1 + 1 \cdot 0 + 1 \cdot (-1) \\ 2 \cdot 1 + (-1) \cdot 0 + 1 \cdot (-1) \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \checkmark \text{ correct}$

* • C16 Method 2: $\nearrow \text{computing } A^{-1}$

Gauss-Jordan row operation:

Augmented-matrix: $[A|I]$, Performing elementary row operations until the left side becomes I , the right side will then be A^{-1}

$A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{bmatrix} \quad [A|I] = \left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 0 \\ 2 & -1 & 1 & 0 & 0 & 1 \end{array} \right]$

1. $R_2 \leftarrow R_2 - R_1, R_3 \leftarrow 2R_1$: $\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & -1 & -1 & -2 & 0 & 1 \end{array} \right]$

2. $R_3 \leftarrow R_3 + R_2$: $\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & -3 & 1 & 1 \end{array} \right]$

3. $R_3 \leftarrow R_3 \leftarrow -R_3$: $\left[\begin{array}{ccc|ccc} 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 3 & -1 & -1 \end{array} \right]$

4. $R_1 \leftarrow R_1 - R_3$: $\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -2 & 1 & 1 \\ 0 & 1 & 0 & -1 & 1 & 0 \\ 0 & 0 & 1 & 3 & -1 & -1 \end{array} \right]$

$A^{-1} = \begin{bmatrix} -2 & 1 & 1 \\ -1 & 1 & 0 \\ 3 & -1 & -1 \end{bmatrix}$

Left block is I_3 . Right block is A^{-1} .

Answer.

check

$\hookrightarrow A \cdot A^{-1}$

$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 2 & -1 & 1 \end{bmatrix} \begin{bmatrix} -2 & 1 & 1 \\ -1 & 1 & 0 \\ 3 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \checkmark \text{ correct}$